

Physical Chemistry (I)

Lecture 18

Deviations from Ideality Phase Diagram of Mixtures



In the Last Class

- Gas mixtures
 - Perfect gas mixture
 - Real gas mixture: fugacity
- Liquid mixtures
 - Ideal solutions
 - Real solutions: ideal-dilute solutions
 - Real solutions: regular solutions (BW model)

Last class

Dalton's Law

Partial pressure for species J is

$$p_J = x_J p$$

so

$$p_A + p_B + \cdots = (x_A + x_B + \cdots)p = p$$

Valid for all gases (not only perfect)

Mixing of Perfect Gases

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

- Entropy of mixing

$$\Delta_{\text{mix}}S = -\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T}\right)_p = -nR(x_A \ln x_A + x_B \ln x_B)$$

- Enthalpy of mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T\Delta_{\text{mix}}S = 0$$

Last class

Real Gases

For a perfect gas,

$$\mu = \mu^\ominus + RT \ln \frac{p}{p^\ominus}$$

For a real gas,

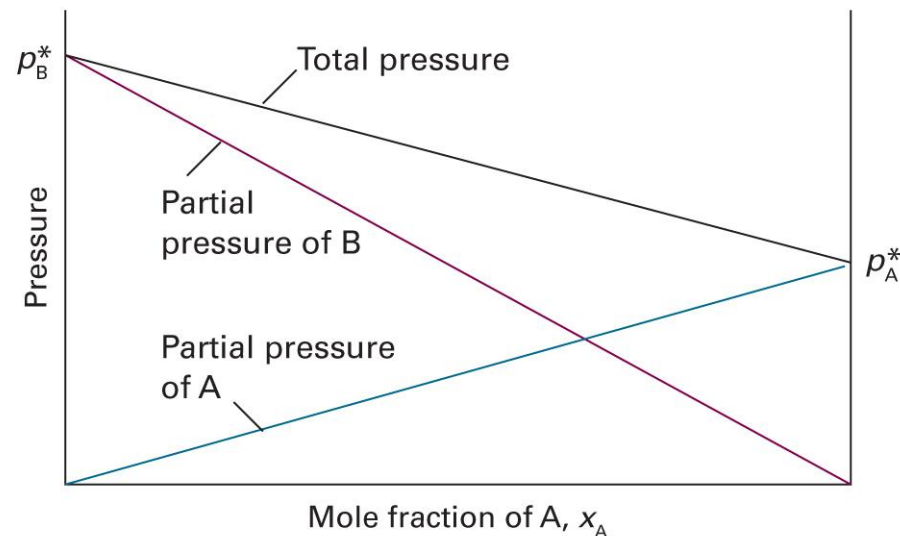
$$\mu = \mu^\ominus + RT \ln \frac{f}{p^\ominus}$$

where f is the **fugacity**.

Raoult's Law: Ideal Solution

$$p_A = x_A p_A^*$$

- **Ideal solutions:** mixtures that obey the law throughout the composition range from pure A to pure B



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.11)

Ideal Solutions

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

- Entropy of mixing

$$\Delta_{\text{mix}}S = -\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T}\right)_p = -nR(x_A \ln x_A + x_B \ln x_B)$$

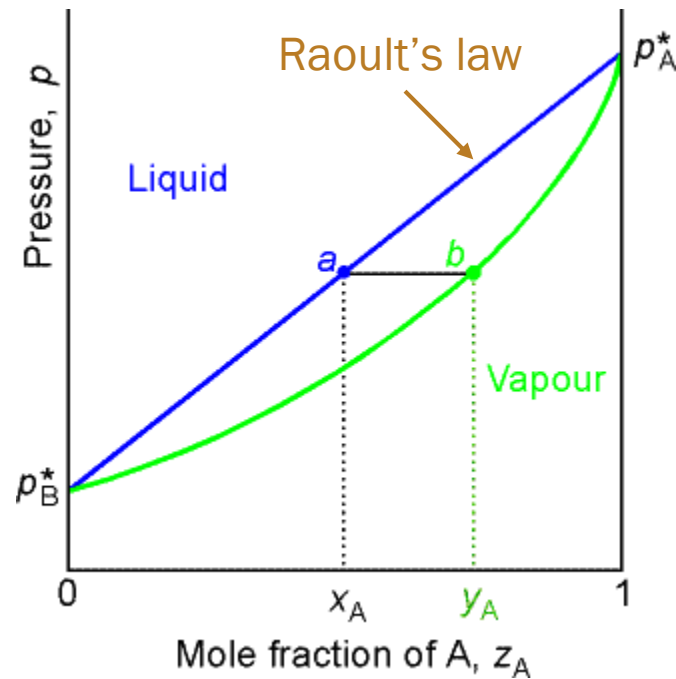
- Enthalpy of mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T\Delta_{\text{mix}}S = 0$$

(Same with perfect gas mixtures)

Last class

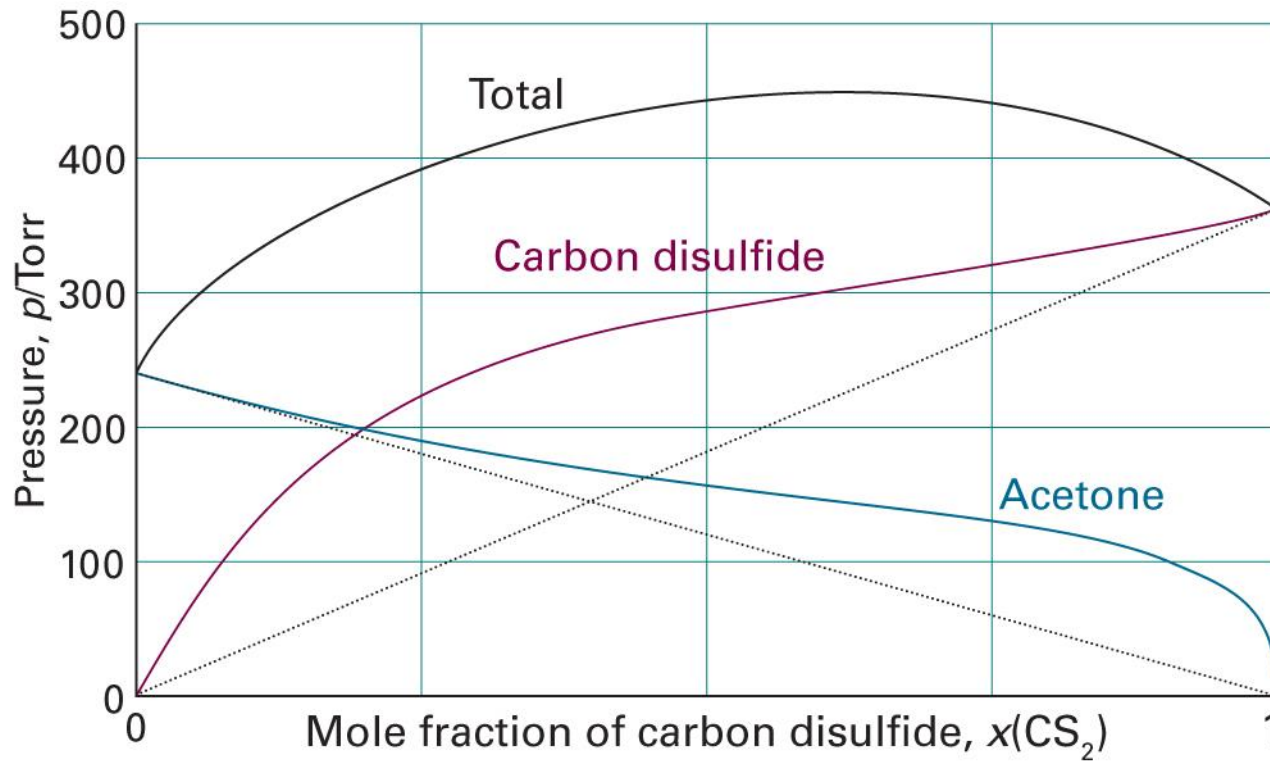
Pressure-Composition Diagram



(Image: <http://people.cst.cmich.edu/teckl1mm/pchemi/chm351ch8af01.htm>)

Last class

Real Solutions: Deviations

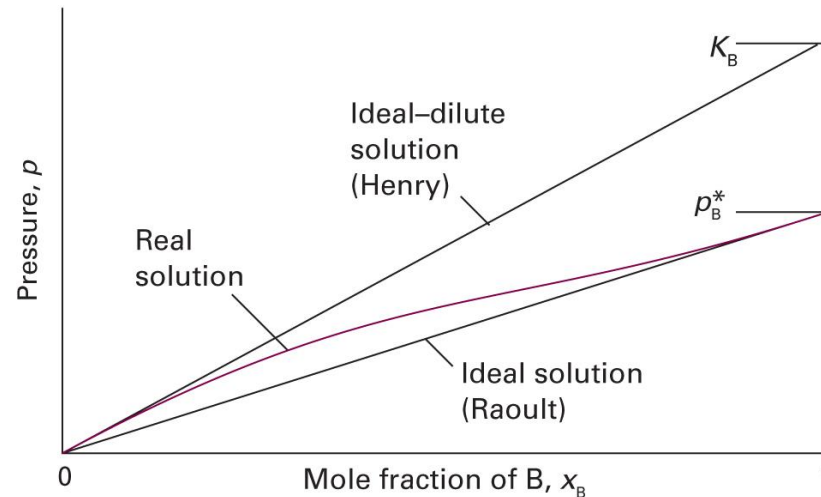


(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.12)

Henry's Law

$$p_B = x_B K_B$$

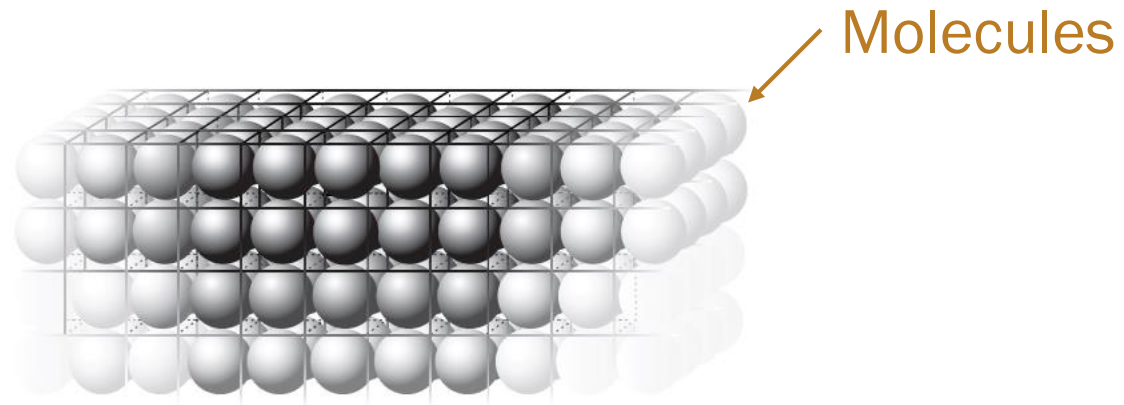
- **Ideal-dilute solutions:** mixtures for which the solute B obeys Henry's law and the solvent A obeys Raoult's law



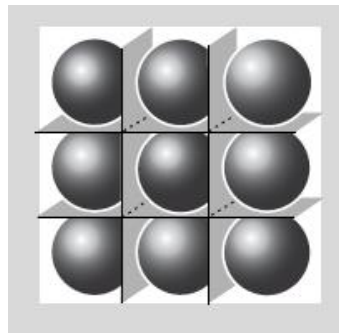
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.13)

Last class

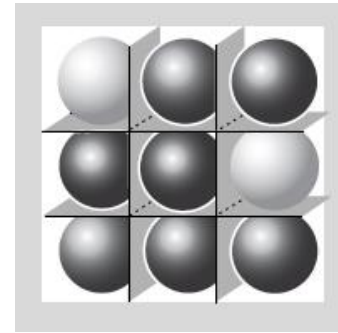
Lattice Model: Bragg-Williams



Pure substance



Mixture



(Image: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figures 14.1 and 16.2)

Last class

Thermodynamics of BW Model

$$\Delta_{\text{mix}}S = -nR(x_A \ln x_A + x_B \ln x_B)$$

$$\Delta_{\text{mix}}U = \xi k_B T \frac{N_A N_B}{N} = \xi N k_B T x_A x_B = \xi n R T x_A x_B$$

$$\Delta_{\text{mix}}A = \Delta_{\text{mix}}U - T \Delta_{\text{mix}}S$$

$$\Delta_{\text{mix}}A = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B)$$

Regular solution: $\Delta_{\text{mix}}S = \Delta_{\text{mix}}S^{\text{ideal}}$ and $\Delta_{\text{mix}}U \neq 0$

Deviations from Ideality

- Excess function

$$\chi^E = \Delta_{\text{mix}}X - \Delta_{\text{mix}}X^{\text{ideal}}$$

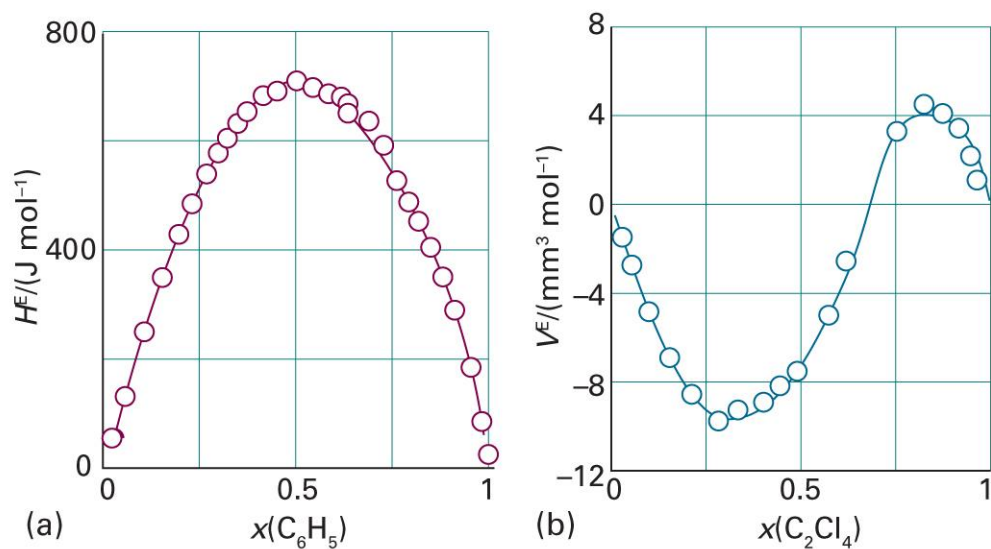
- Activity

$$\mu_A = \mu_A^* + RT \ln a_A$$

Excess Function

$$X^E = \Delta_{\text{mix}}X - \Delta_{\text{mix}}X^{\text{ideal}}$$

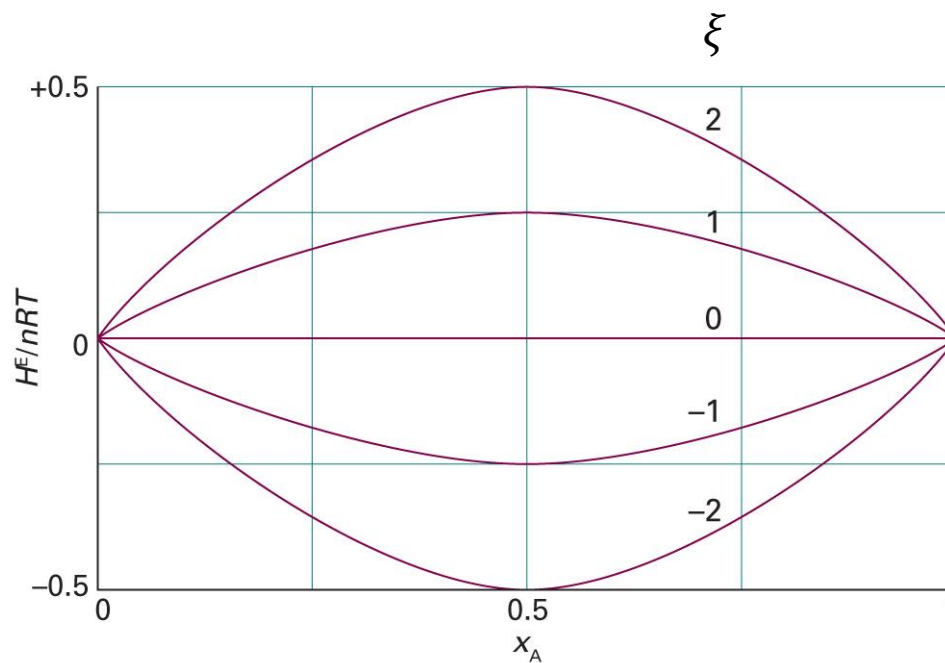
- Regular solution: $S^E = 0$ and $U^E \neq 0$ (or $H^E \neq 0$)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.3)

BW Model: Excess Enthalpy

$$H^E = \xi nRT x_A x_B$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.4)

Activity a_J

- Ideal-dilute solution

- Solvent: $\mu_A(l) = \mu_A^*(l) + RT \ln x_A$

- Solute: $\mu_B(l) = \mu_B^\ominus(l) + RT \ln x_B$

where $\mu_B^\ominus(l) = \mu_B^*(l) + RT \ln(K_B/p_B^*)$

- Real solution

- Solvent: $\mu_A(l) = \mu_A^*(l) + RT \ln a_A$

- Solute: $\mu_B(l) = \mu_B^\ominus(l) + RT \ln a_B$

Measurement of Activity

$$\mu(l) = \mu^*(l) + RT \ln(p/p^*)$$

- Solvent

$$\mu_A(l) = \mu_A^*(l) + RT \ln a_A$$

$$a_A = p_A/p_A^*$$

- Solute

$$\mu_B(l) = \mu_B^\ominus(l) + RT \ln a_B = \mu_B^*(l) + RT \ln(K_B a_B/p_B^*)$$

$$a_B = p_B/K_B$$

Activity Coefficient

$$\gamma_J = a_J/x_J$$

- Solvent

$$\mu_A(l) = \mu_A^*(l) + RT \ln a_A = \mu_A^*(l) + RT \ln x_A + RT \ln \gamma_A$$

- Solute

$$\mu_B(l) = \mu_B^\ominus(l) + RT \ln a_B = \mu_B^\ominus(l) + RT \ln x_B + RT \ln \gamma_B$$

As $x_A \rightarrow 1$ and $x_B \rightarrow 0$, $a_J \rightarrow x_J$ and $\gamma_J \rightarrow 1$

Activity of BW Model

$$\begin{aligned}\Delta_{\text{mix}}G &= G - G^* = n_A\mu_A + n_B\mu_B - n_A\mu_A^* - n_B\mu_B^* \\ &= n_A(\mu_A - \mu_A^*) + n_B(\mu_B - \mu_B^*) \\ &= n_ART \ln a_A + n_BRT \ln a_B \\ &= nRT(x_A \ln a_A + x_B \ln a_B) \\ &= nRT(x_A \ln x_A + x_B \ln x_B + x_A \ln \gamma_A + x_B \ln \gamma_B)\end{aligned}$$

Comparing with the BW model Gibbs energy of mixing,

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B),$$

$$x_A \ln \gamma_A + x_B \ln \gamma_B = \xi x_A x_B$$

Activity of BW Model

$$x_A \ln \gamma_A + x_B \ln \gamma_B = \xi x_A x_B$$

$$\xi x_A x_B = \xi x_A x_B (x_A + x_B) = x_A (\xi x_B^2) + x_B (\xi x_A^2)$$

Thus, the following equations satisfy the given criterion:

$$\ln \gamma_A = \xi x_B^2, \text{ or } \gamma_A = e^{\xi x_B^2}$$

$$\ln \gamma_B = \xi x_A^2, \text{ or } \gamma_B = e^{\xi x_A^2}$$

(Margules equations)

BW Model and the Two Laws

$$\gamma_A = e^{\xi x_B^2}, \quad \gamma_B = e^{\xi x_A^2}$$

For component A, the activity is

$$a_A = x_A \gamma_A = x_A e^{\xi x_B^2} = x_A e^{\xi(1-x_A)^2}$$

From the property of the activity, $a_A = p_A/p_A^*$ and

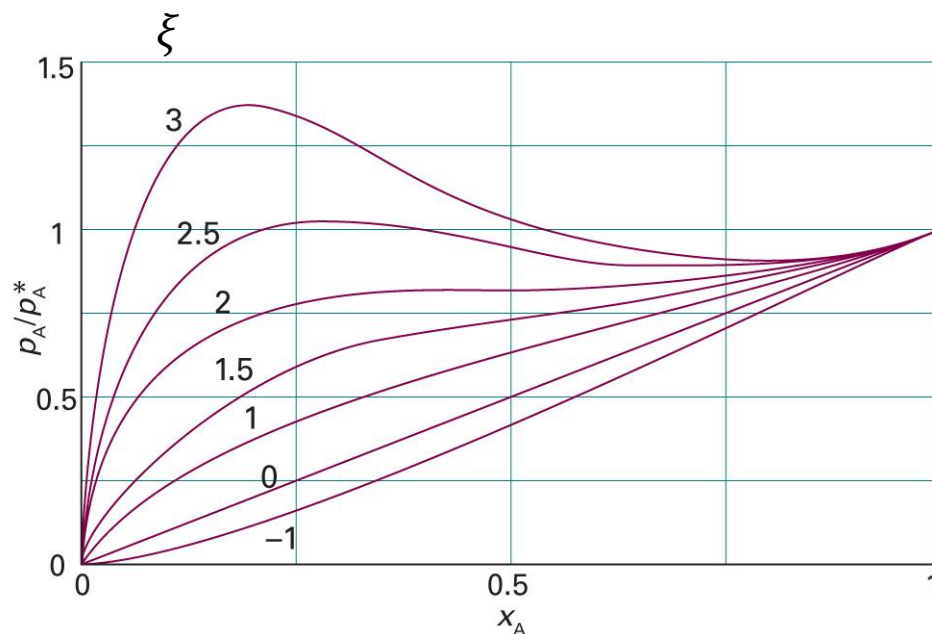
$$p_A = p_A^* a_A = p_A^* x_A e^{\xi(1-x_A)^2}$$

- $x_A \rightarrow 1$: $p_A = p_A^* x_A e^0 = p_A^* x_A$ (Raoult's law)
- $x_A \rightarrow 0$: $p_A = p_A^* x_A e^{\xi} = K_A x_A$ (Henry's law)

$$\text{where } K_A = p_A^* e^{\xi}$$

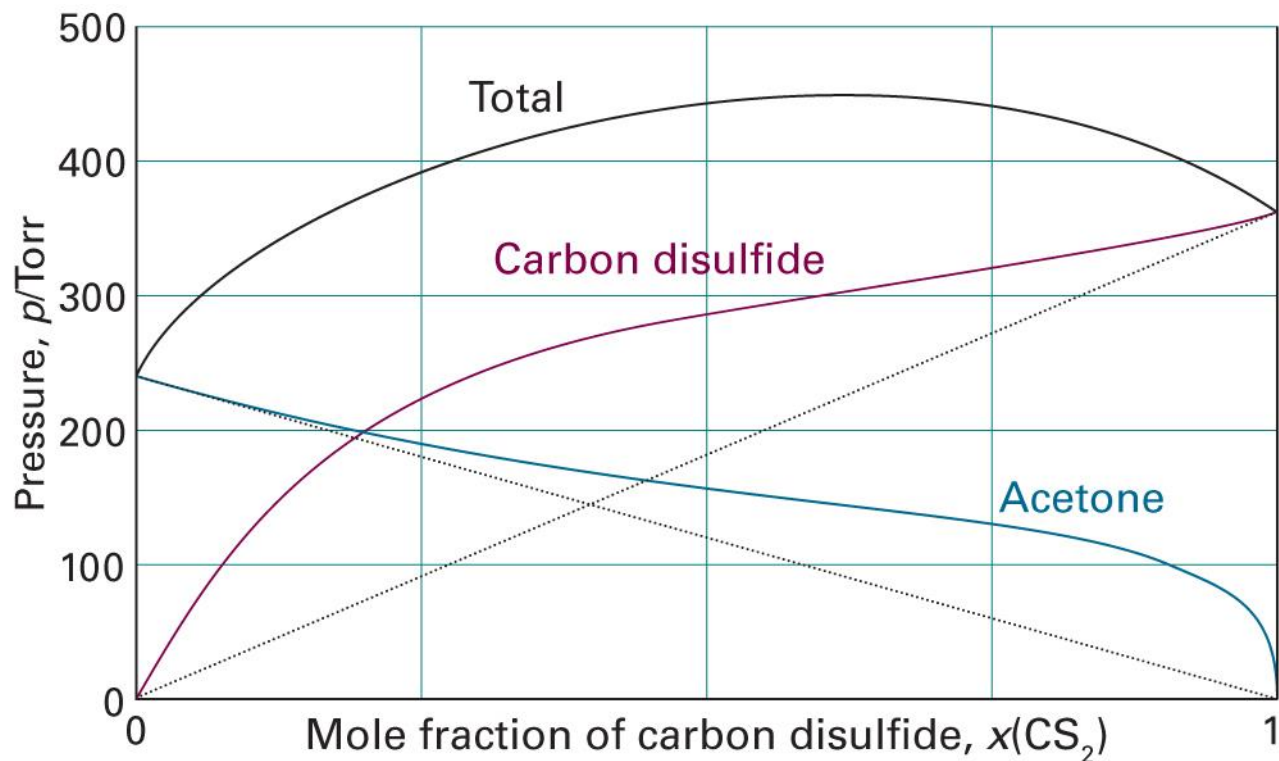
Vapor Pressure of BW Model

$$p_A/p_A^* = x_A e^{\xi(1-x_A)^2}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5F.2)

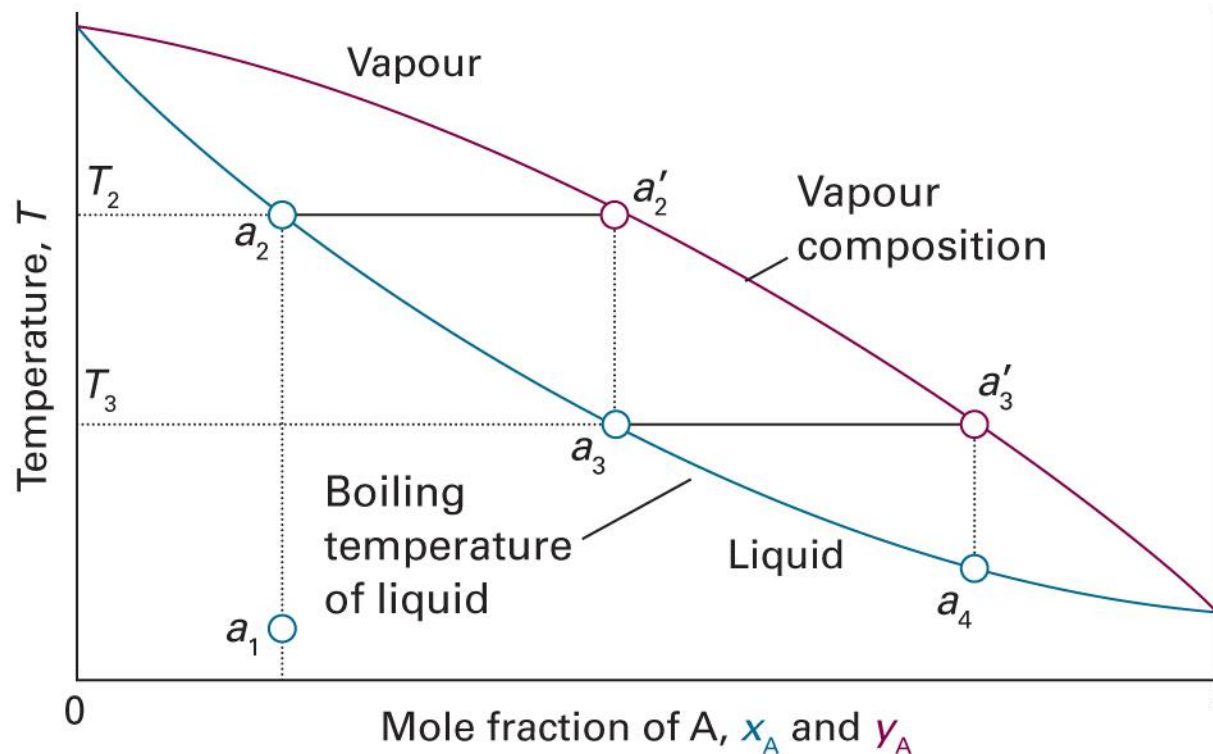
Now We Can Explain This



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.12)

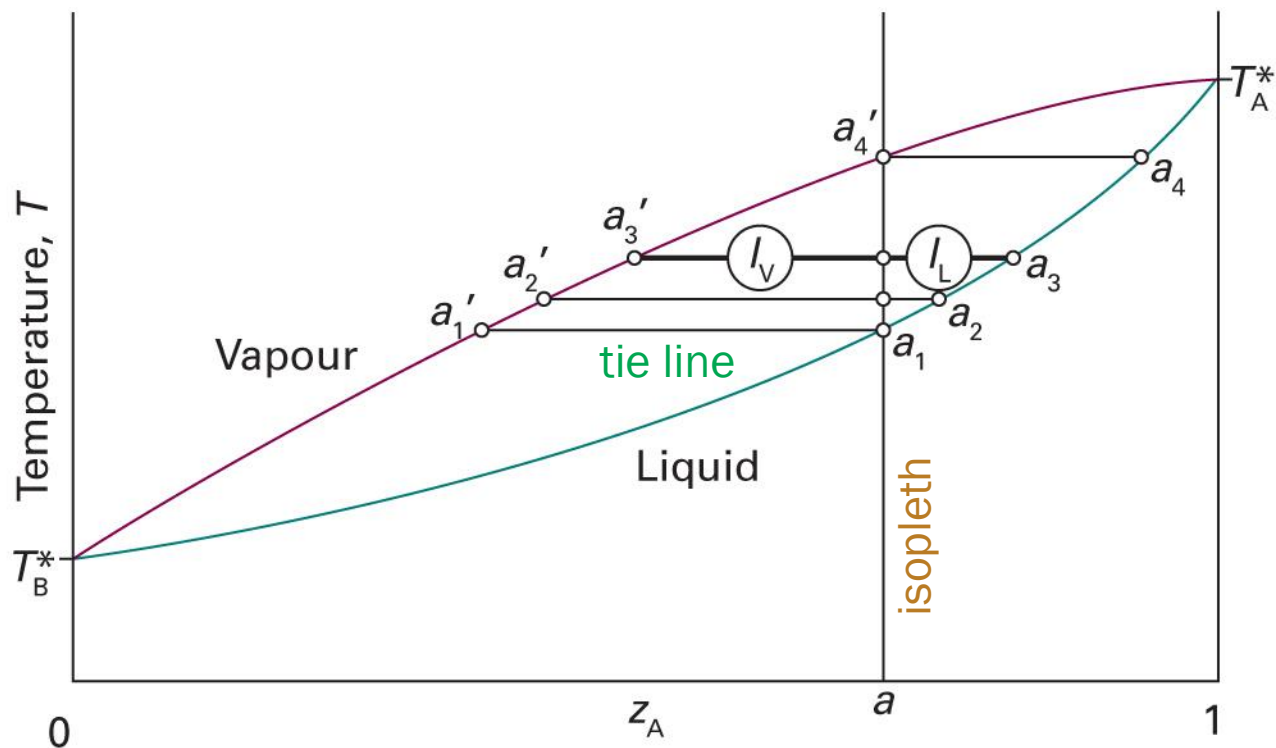
Phase Diagram: Ideal Solutions

$$p_A^* > p_B^* \text{ (A is more volatile)}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.4)

Interpretation



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.7)

Lever Rule

- Total amount of A molecules: $n_A = n_{A,l} + n_{A,g}$
- Total amount of molecules in liquid: $n_l = n_{A,l} + n_{B,l}$
- Total amount of molecules in gas: $n_g = n_{A,g} + n_{B,g}$

$$n_{A,l} = x_A n_l, \quad n_{A,g} = y_A n_g$$

Hence,

$$n_A = x_A n_l + y_A n_g$$

Lever Rule

$$n_A = x_A n_l + y_A n_g$$

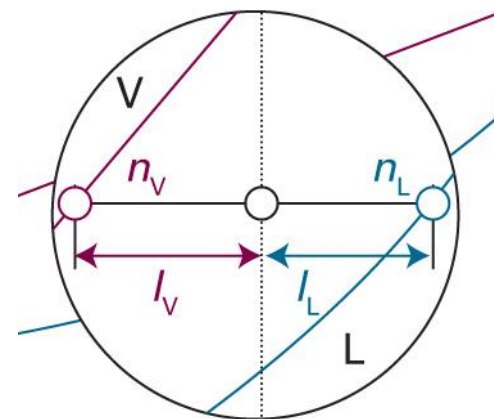
On the other hand,

$$n_A = n z_A = z_A n_l + z_A n_g$$

Equating the two equations,

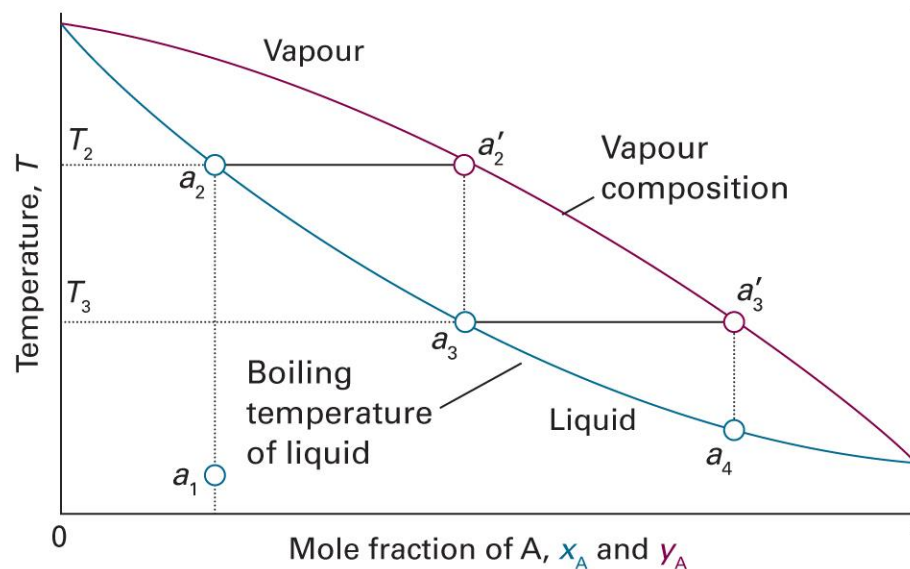
$$n_l(z_A - x_A) = n_g(y_A - z_A)$$

$$n_l l_l = n_g l_g$$



Phase Diagrams of Real Solutions

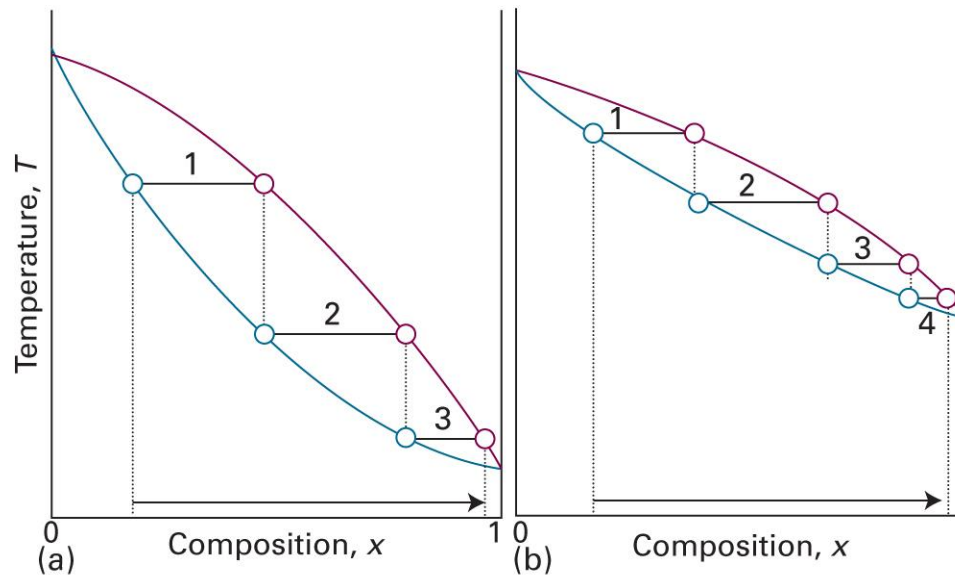
- Liquid-liquid equilibrium: phase separation can occur
- Liquid-vapor equilibrium: deviations from ideal behavior



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.4)

Distillation

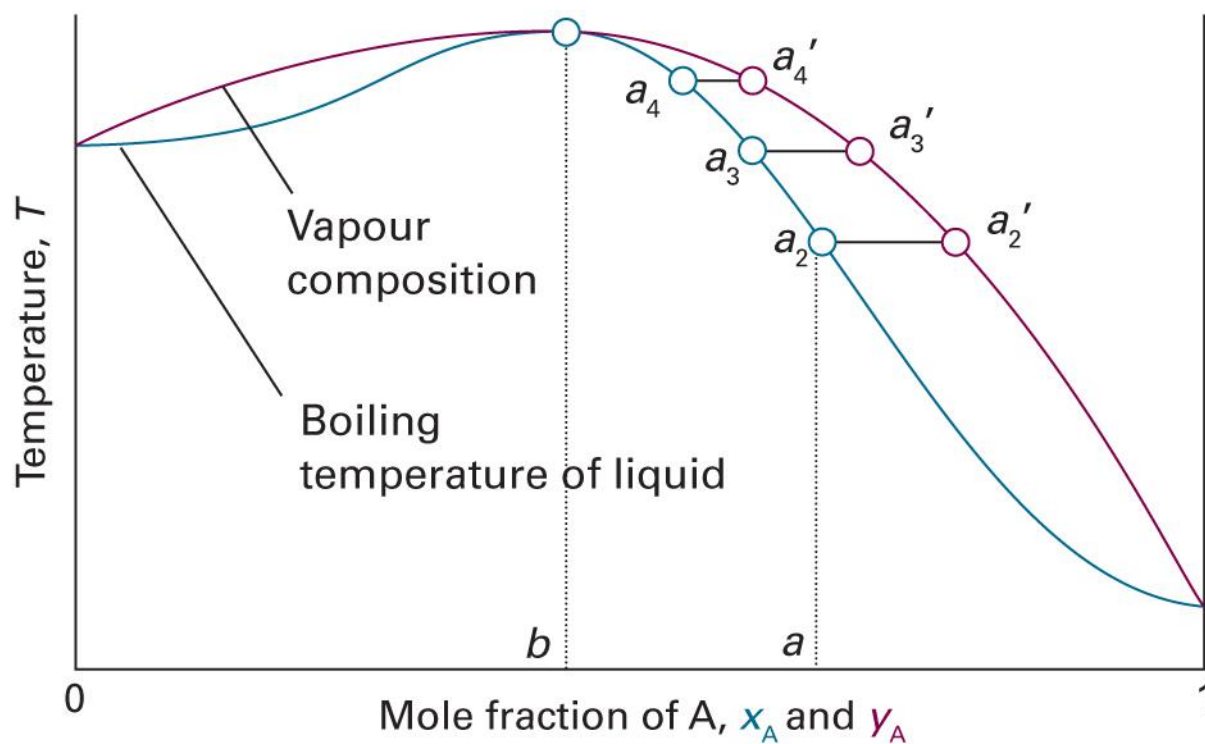
- **Simple distillation:** vapor withdrawn and condensed
- **Fractional distillation:** the cycle repeated successively



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.9)

High-Boiling Azeotrope

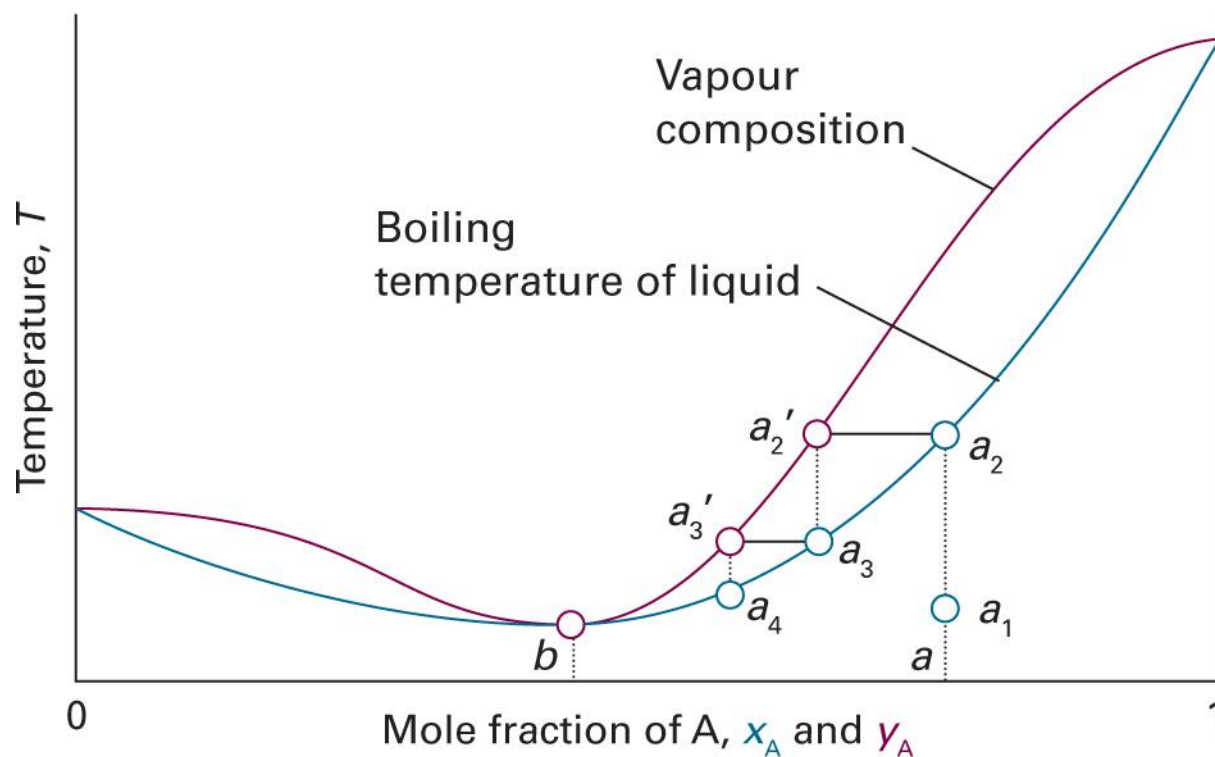
$G^E < 0$ (A-B interactions stronger)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.10)

Low-Boiling Azeotrope

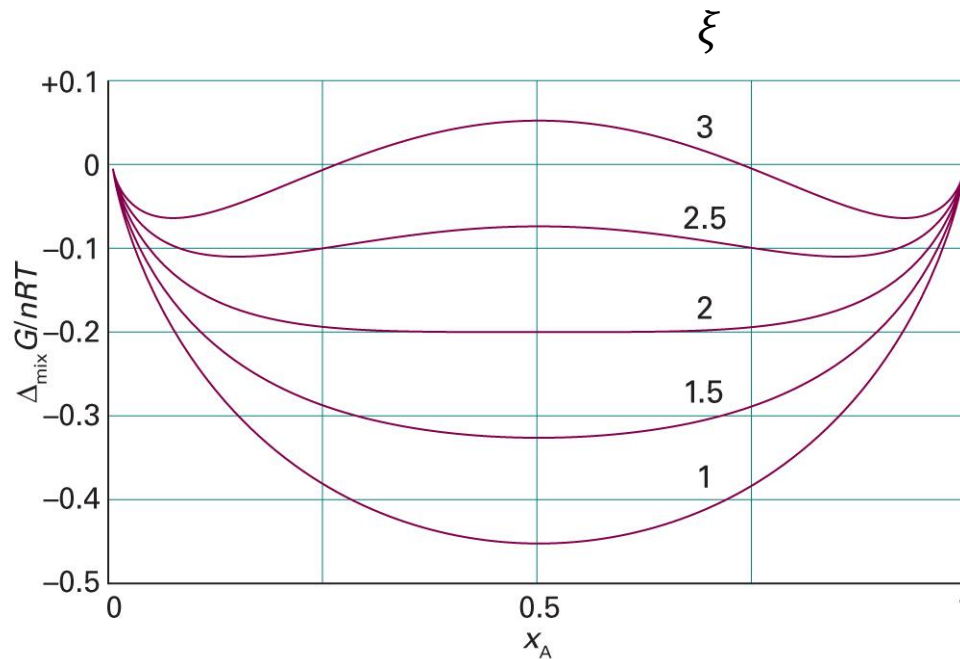
$G^E > 0$ (A-B interactions weaker)



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 5C.11)

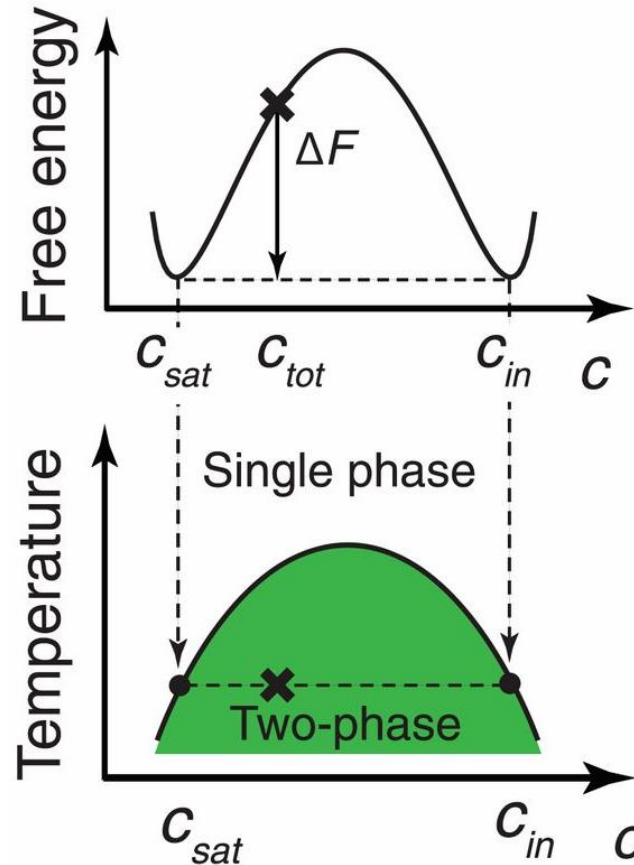
BW Model: Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.5)

What Happens at $\xi > 2$?



(Image: Shin and Brangwynne, *Science* 357: eaaf4382 (2017), Figure 3)

Locations of Minima

$$\begin{aligned}\Delta_{\text{mix}}G &= nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B) \\ &= nRT\{x_A \ln x_A + (1 - x_A) \ln(1 - x_A) + \xi x_A(1 - x_A)\}\end{aligned}$$

The local minima can be found by using the derivative:

$$\begin{aligned}&\left(\frac{\partial \Delta_{\text{mix}}G}{\partial x_A}\right)_{T,p} \\ &= nRT\{\ln x_A + 1 - \ln(1 - x_A) - 1 + \xi(1 - 2x_A)\} \\ &= nRT\left\{\ln \frac{x_A}{1 - x_A} + \xi(1 - 2x_A)\right\} = 0\end{aligned}$$

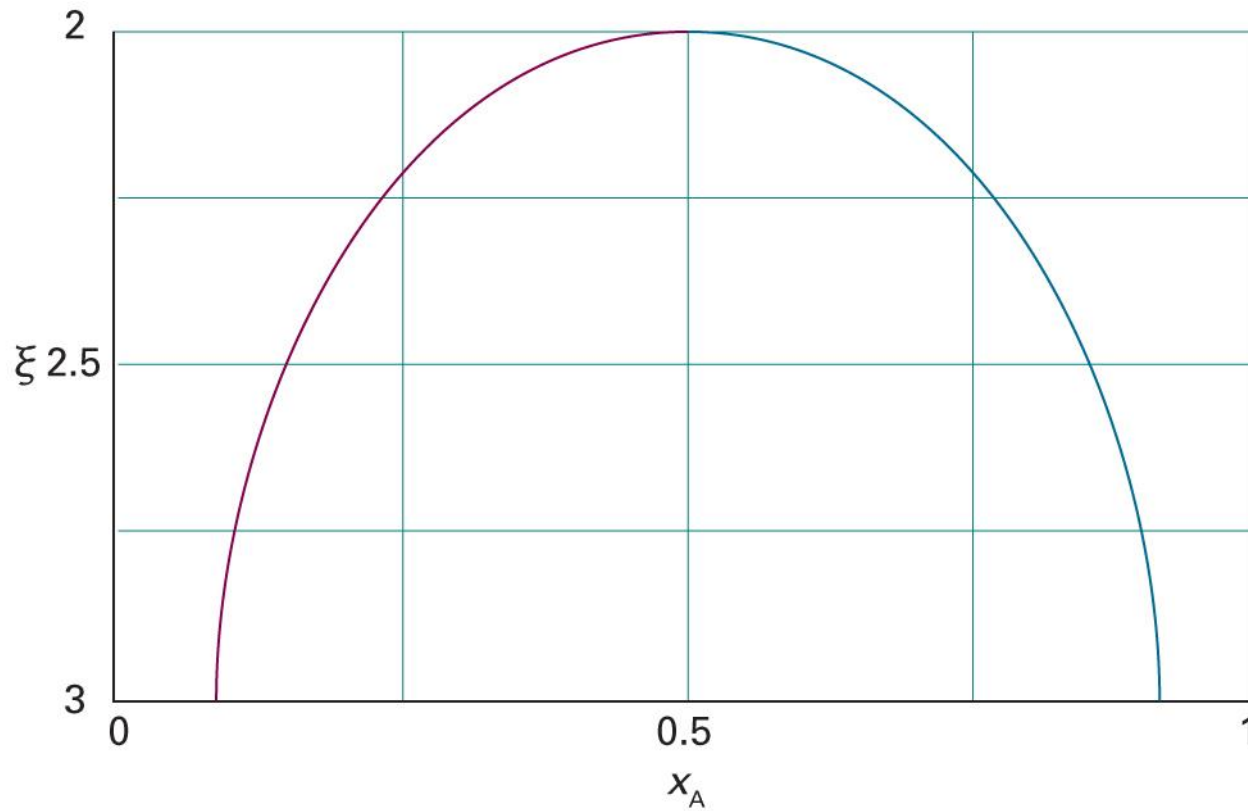
Locations of Minima

$$nRT \left\{ \ln \frac{x_A}{1 - x_A} + \xi(1 - 2x_A) \right\} = 0$$

The minima are located at x_A such that

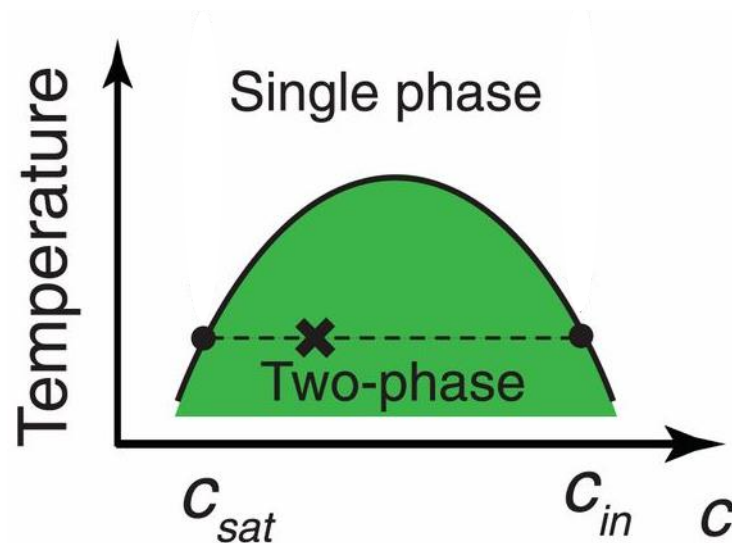
$$\ln \frac{x_A}{1 - x_A} = -\xi(1 - 2x_A)$$

Locations of Minima



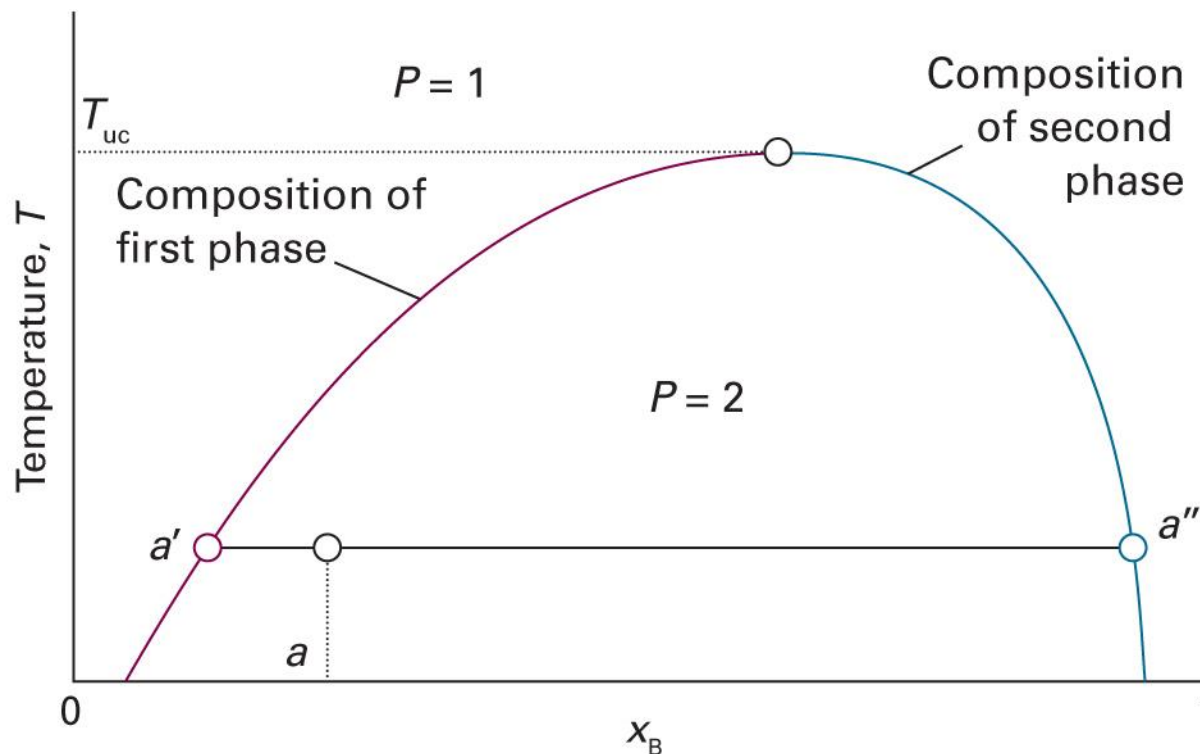
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.17)

Phase Diagram



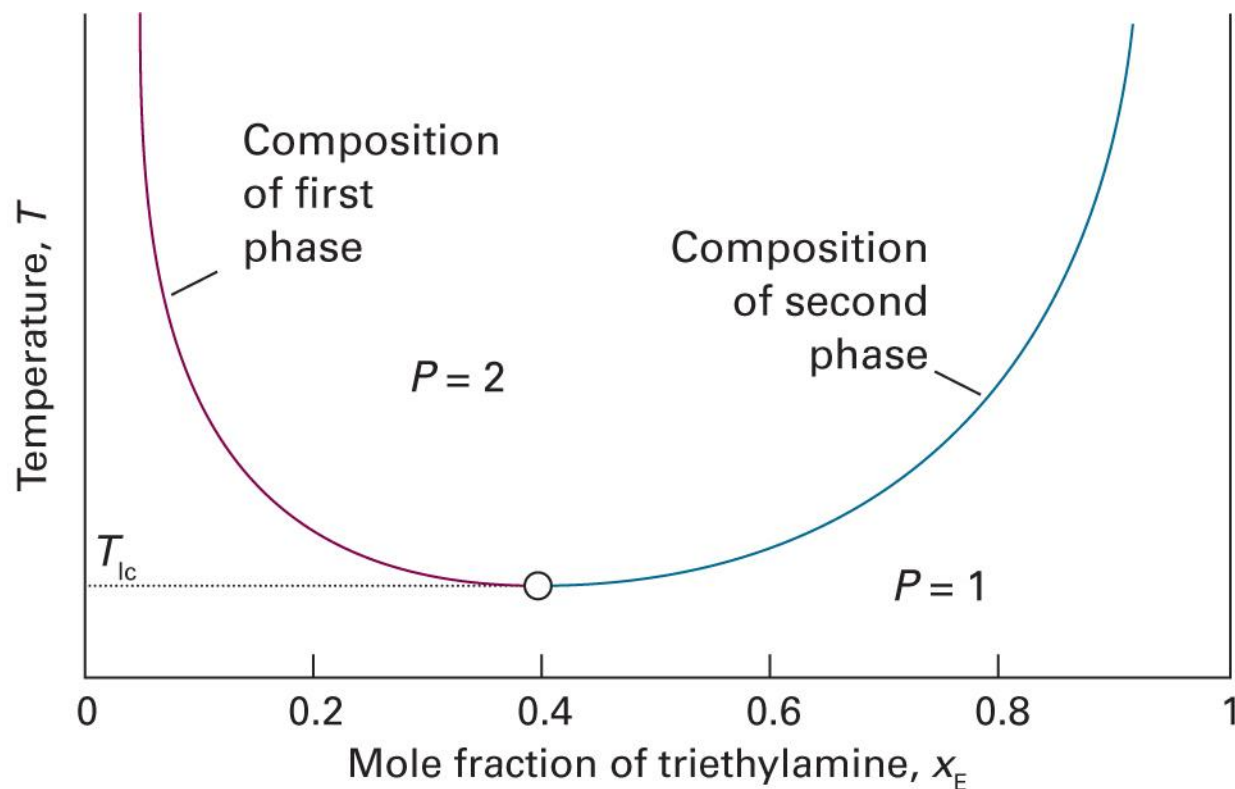
Lever rule applies in the two-phase region!

Upper Critical Solution Temperature

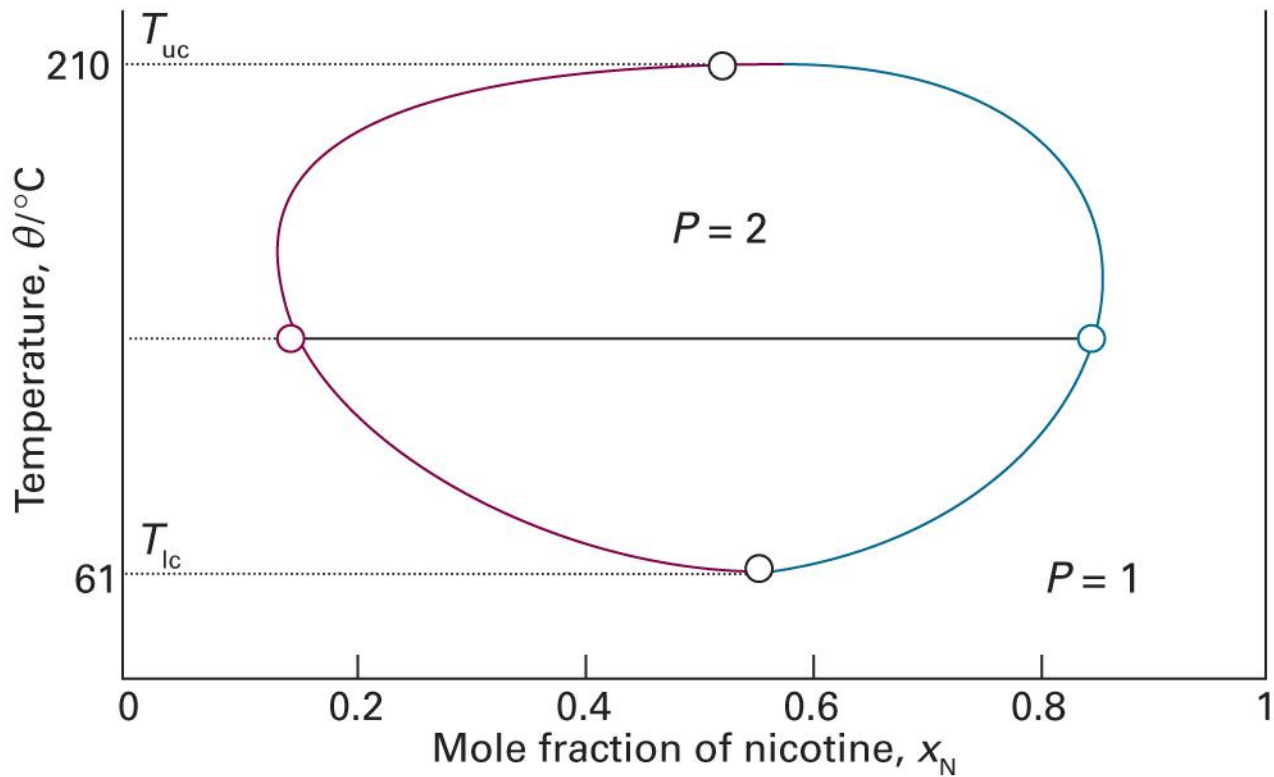


(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.13)

Lower Critical Solution Temperature



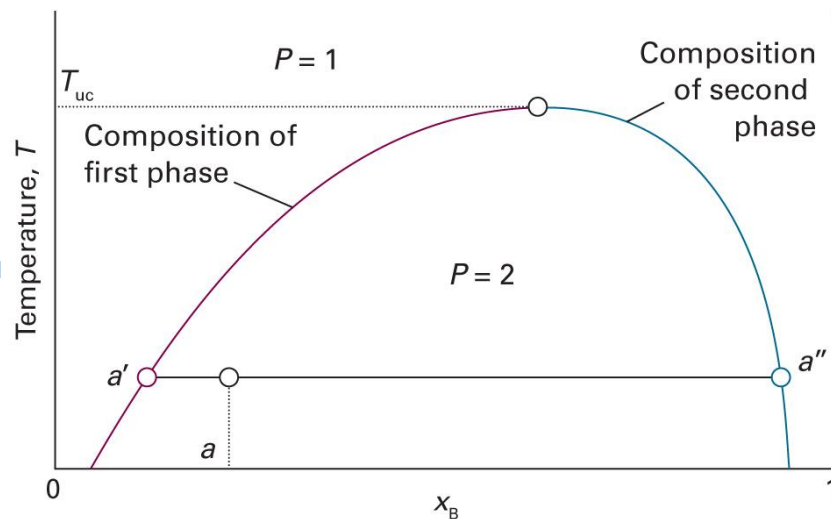
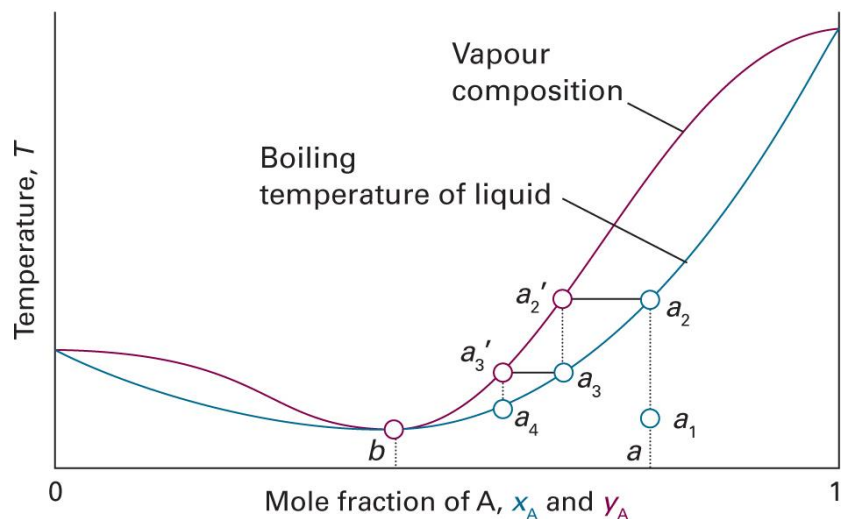
Both UCST and LCST



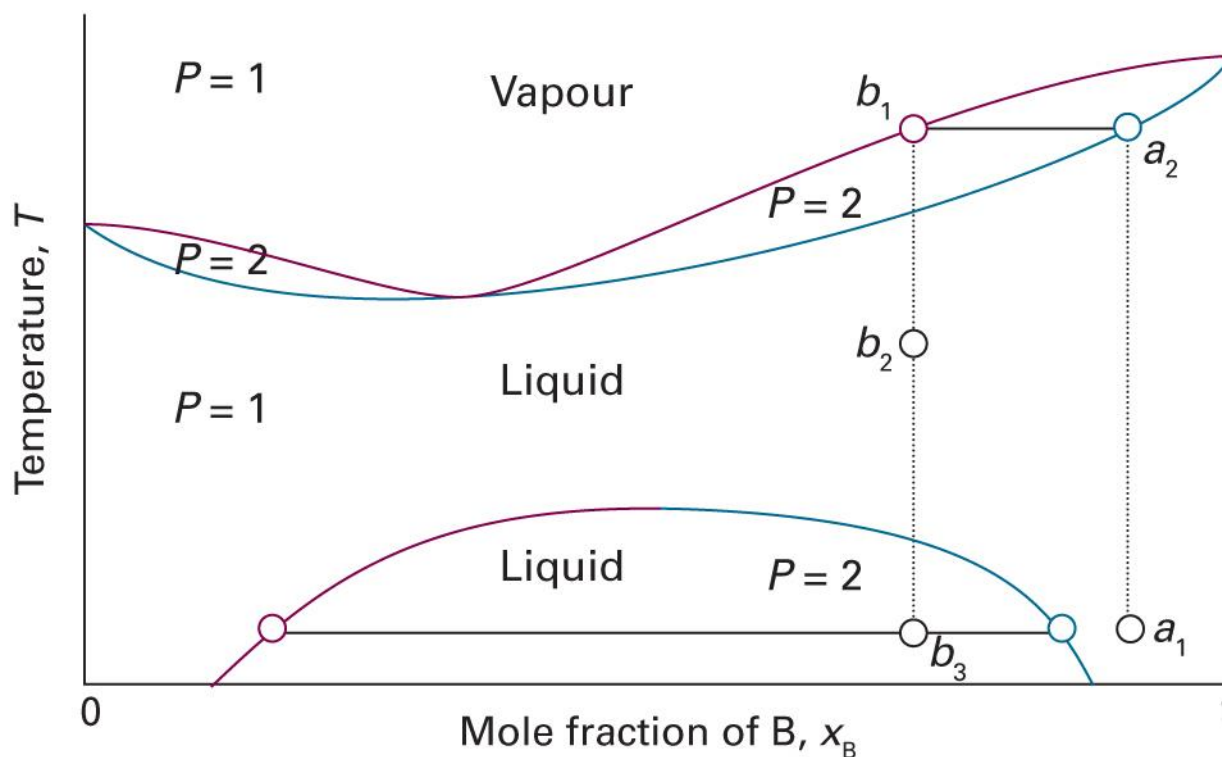
(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 5C.19)

L-G and L-L Equilibria

When A-B interactions are weaker than A-A and B-B

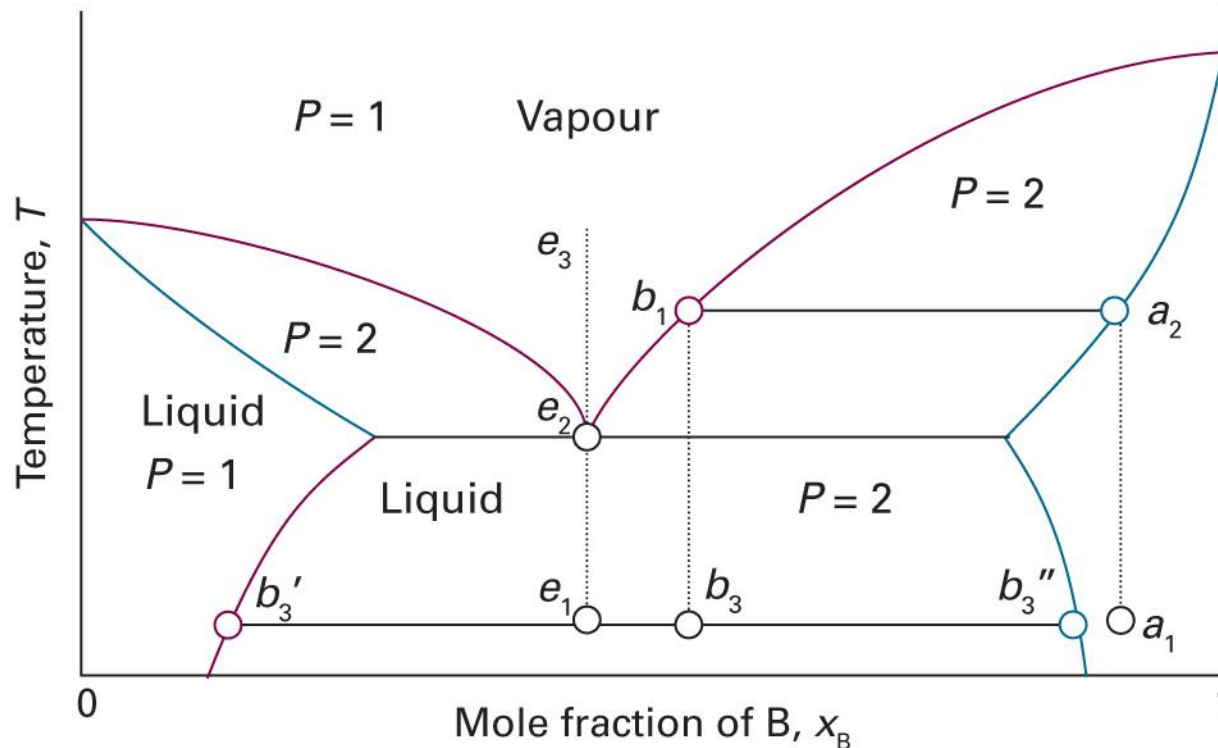


Case 1: Miscible before Boiling



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.20)

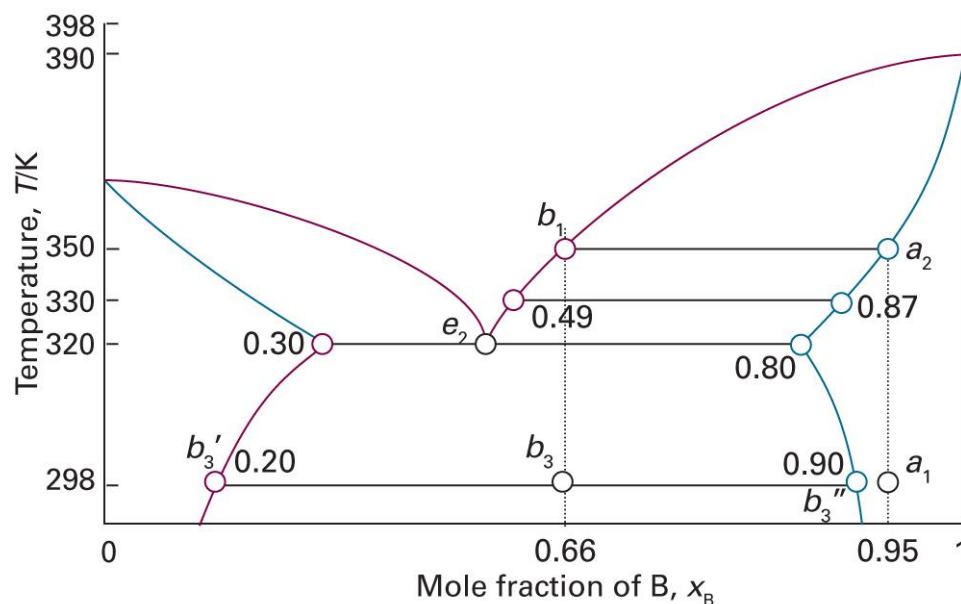
Case 2: Boiling before Miscible



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.21)

Example 5C.3

State the changes that occur when a mixture of composition $x_B = 0.95$ (a_1) in the figure below is boiled and the vapour condensed.



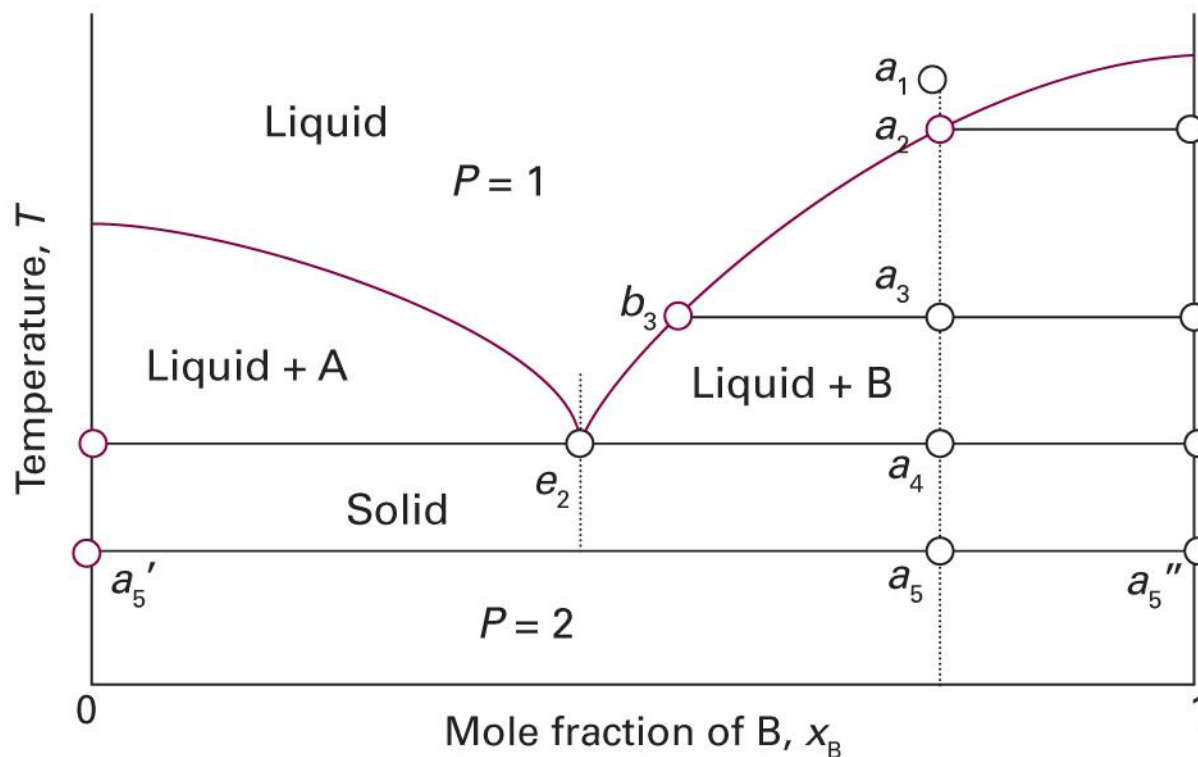
Liquid-Solid Equilibrium

Consider a two-component system (A and B)
where the two solids are almost immiscible.

Upon mixing,

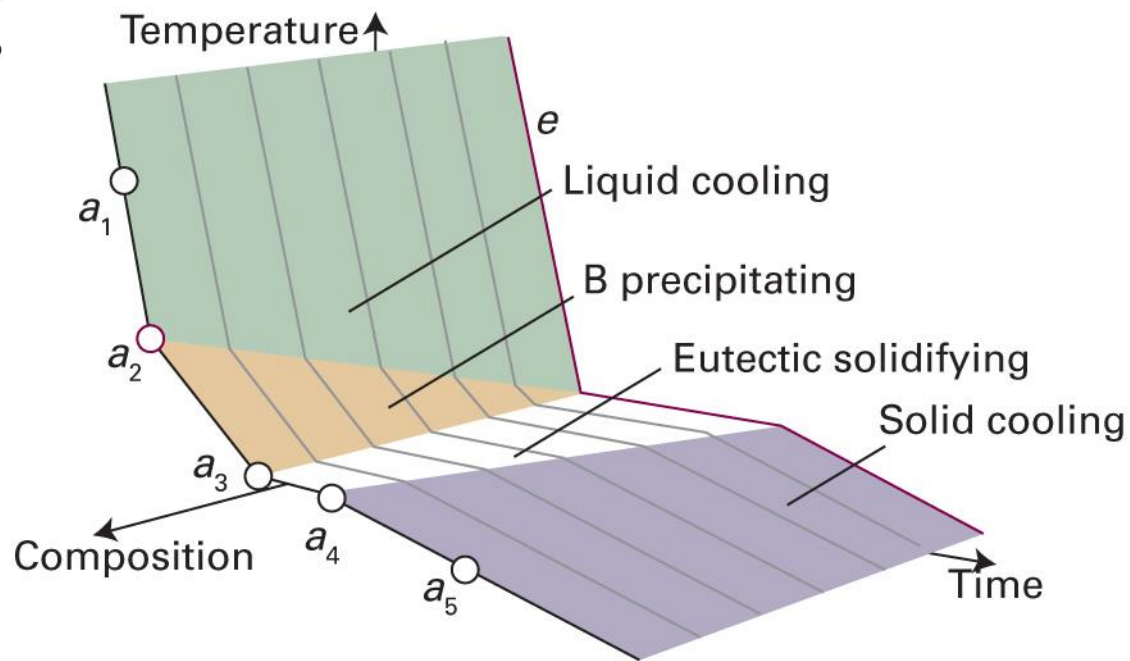
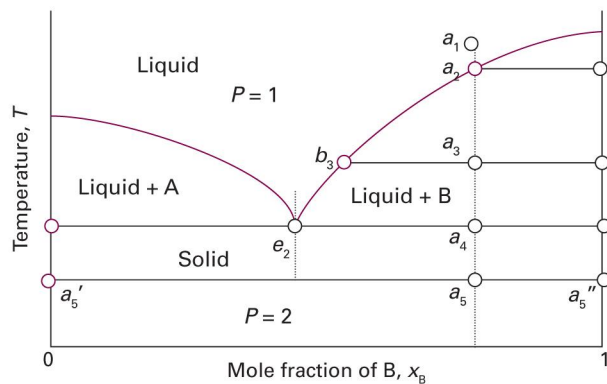
- Melting temperature ↓: A-B interactions are weaker
- Melting temperature ↑: A-B interactions are stronger

Phase Diagrams Involving Solids



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5D.1)

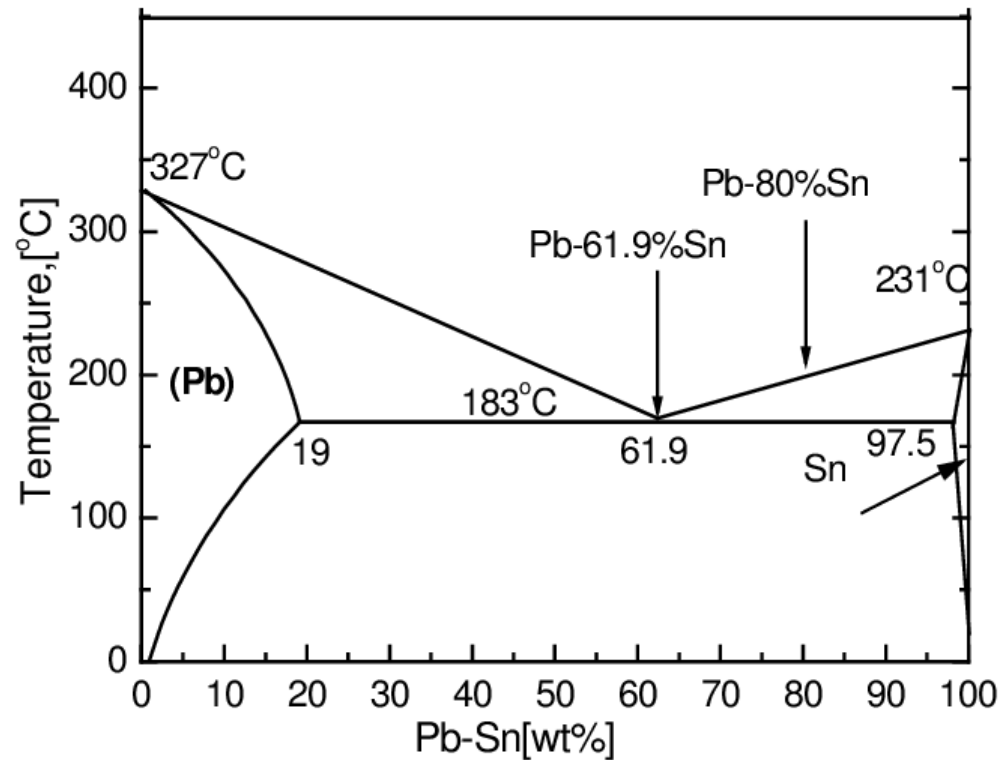
Eutectics



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 5D.1 and 5D.2)

Not in Atkins

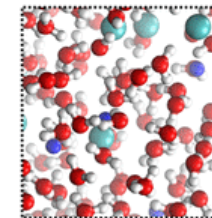
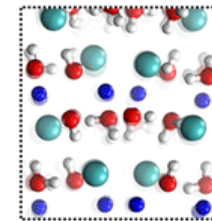
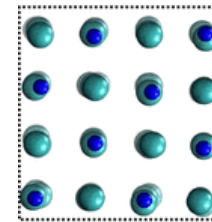
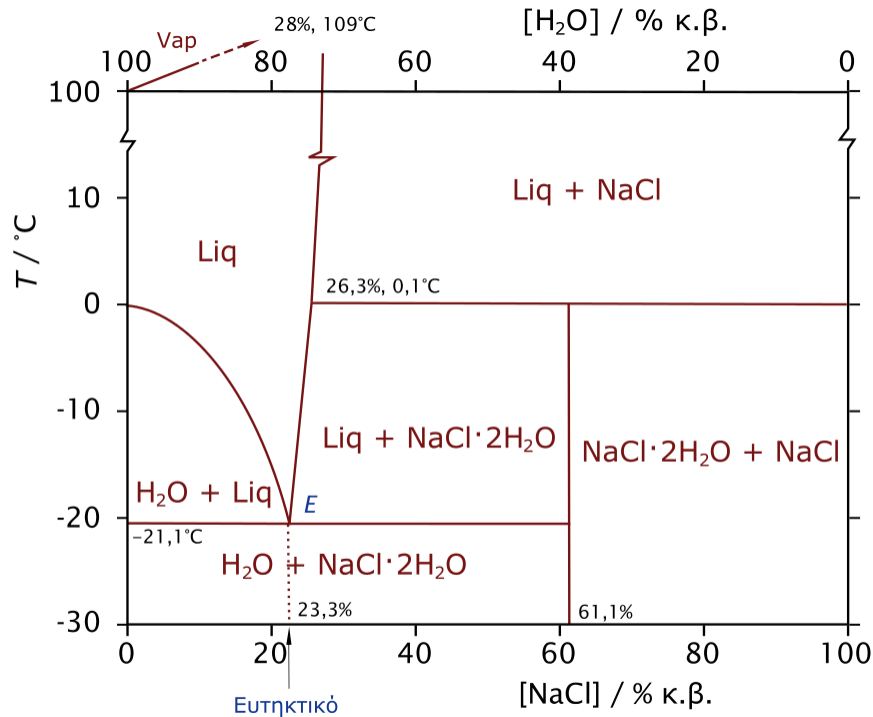
Pb-Sn System



(Image: https://www.researchgate.net/figure/The-Pb-Sn-phase-diagram_fig1_291162603)

Not in Atkins

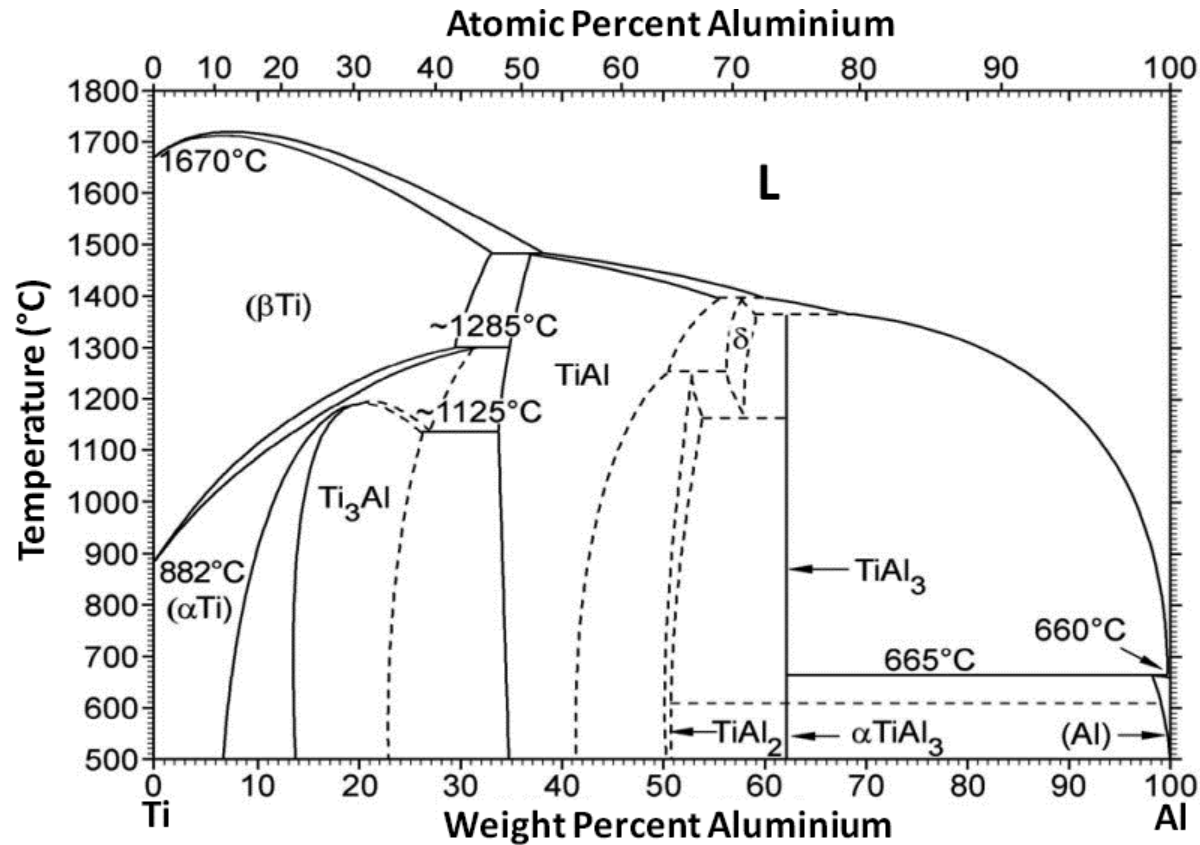
H₂O-NaCl System



(Image: <https://commons.wikimedia.org/wiki/File:H2O-NaCl-phase-diagram-greek.svg>)

Not in Atkins

Ti-Al System



(Image: Mitra and Wanhill, *Structural Intermetallics* (2016), Figure 10)